

Modeling Data

Dominguez Center for Data Science

2026-05-29

Questions Round-Up

As we wrap up...

- I'd recommend downloading R and RStudio to your personal computer. You can find directions [here](#).
- Make sure to export all your materials from Posit Cloud in the next week or so. I will delete the materials on June 30th.
- Attend the weekly **Writing/Coding Sessions**: 2:00 - 4:00pm on Mondays in June & July, Taylor Hall 203.

Models/Tests we are going to cover:

- For modeling a quantitative variable:
 - When you have a single categorical explanatory variable (with two categories): Two sample t-test
 - When you have 1 or more explanatory variables (categorical and/or quantitative): Linear regression
- For modeling a categorical variable:
 - When you have a single categorical explanatory variable: Chi-squared test
 - When you have 1 or more explanatory variables (categorical and/or quantitative): Logistic regression and classification trees

For the sake of time today, not going to focus on model assumptions, but make sure to check them!

Load Packages

```
# For data wrangling and plotting
library(tidyverse)

# For fitting classification trees
library(rpart)

# For creating charts of trees
library(rpart.plot)
```

Data Example:

“The social contract of Halloween is simple: Provide adequate treats to costumed masses, or be prepared for late-night tricks from those dissatisfied with your offer. To help you avoid that type of vengeance, and to help you make good decisions at the supermarket this weekend, we wanted to figure out what Halloween candy people most prefer. So we devised an experiment: Pit dozens of fun-sized candy varieties against one another, and let the wisdom of the crowd decide which one was best.” – Walt Hickey

“While we don’t know who exactly voted, we do know this: 8,371 different IP addresses voted on about 269,000 randomly generated matchups.² So, not a scientific survey or anything, but a good sample of what candy people like.”

Which would you prefer as a trick-or-treater?

Battle: : Candy

Hershey's Special Dark



Payday



```
candy <- read_csv("https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power")
mutate(pricepercent = pricepercent*100)

glimpse(candy)
```

Rows: 85

Columns: 13

```
$ competitorname    <chr> "100 Grand", "3 Musketeers", "One dime", "One quarter~
$ chocolate         <dbl> 1, 1, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~
$ fruity            <dbl> 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, ~
$ caramel           <dbl> 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, ~
$ peanutyalmondy    <dbl> 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ nougat            <dbl> 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~
$ crispedricewafer  <dbl> 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ hard              <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, ~
$ bar               <dbl> 1, 1, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~
$ pluribus          <dbl> 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 1, 1, 0, 1, 0, ~
$ sugarpercent      <dbl> 0.732, 0.604, 0.011, 0.011, 0.906, 0.465, 0.604, 0.31~
$ pricepercent       <dbl> 86.0, 51.1, 11.6, 51.1, 51.1, 76.7, 76.7, 51.1, 32.5,~
$ winpercent         <dbl> 66.97173, 67.60294, 32.26109, 46.11650, 52.34146, 50.~
```

T-test

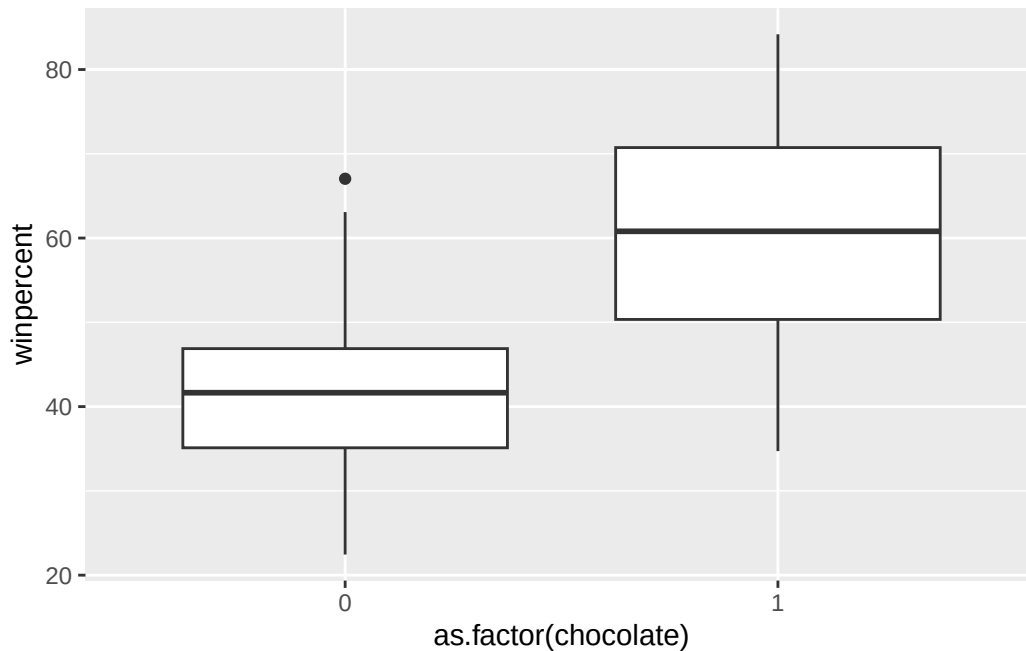
Consider this test when:

- Response variable (y): quantitative
- Explanatory variable (x): categorical with two categories

Let's say we want to predict the win percentage. Which of the categorical variables do you think would be a good predictor of the average win percentage?

First graph the data.

```
ggplot(data = candy, mapping = aes(x = as.factor(chocolate), y = winpercent)) +
  geom_boxplot()
```



Then run the two-sample t-test.

```
t.test(winpercent ~ chocolate, data = candy)
```

Welch Two Sample t-test

data: winpercent by chocolate

t = -7.3031, df = 67.539, p-value = 4.164e-10

alternative hypothesis: true difference in means between group 0 and group 1 is not equal to

95 percent confidence interval:

-23.91110 -13.64744

sample estimates:

mean in group 0 mean in group 1

42.14226 60.92153

Is the variable we selected a good predictor of the average price?

Multiple Linear Regression

Consider this model when:

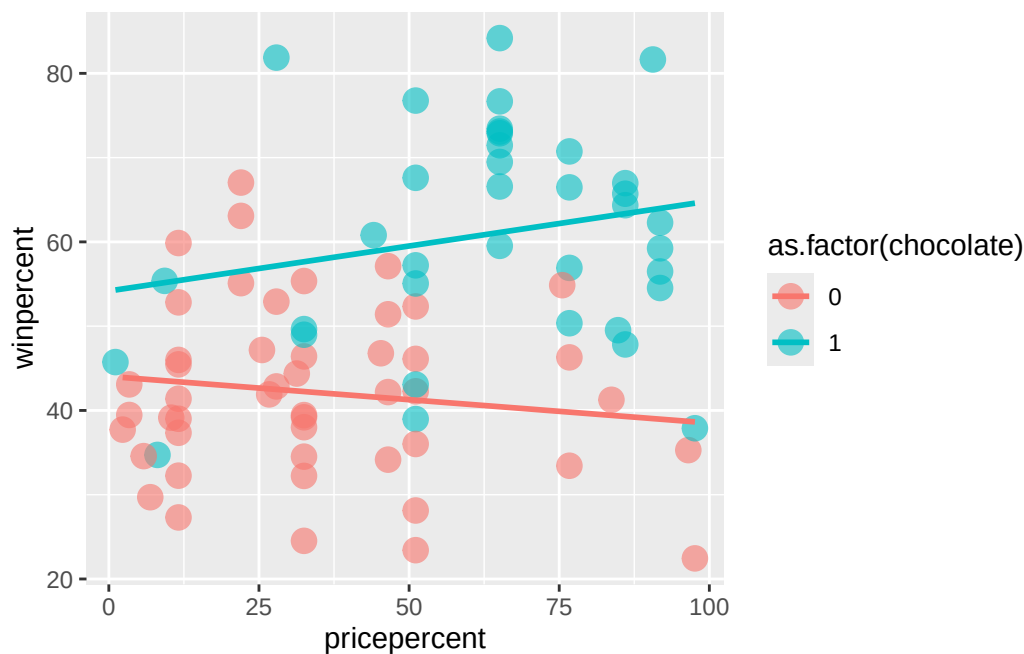
- Response variable (y): quantitative
- Explanatory variable(s) (x)'s: quantitative and/or categorical (simple = 1 explanatory variable)

And the relationship between the x 's and y is reasonably approximated by the regression equation:

$$y = \beta_o + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \epsilon$$

Let's now build a model for win percentage using price percentage and whether or not the candy contains chocolate. But first, we should graph the data.

```
ggplot(data = candy,
       mapping = aes(x = pricepercent,
                     y = winpercent, color = as.factor(chocolate))) +
  geom_point(alpha = 0.6, size = 4) +
  geom_smooth(method = "lm", se = FALSE)
```



Equal slopes model

You need to first determine if you want to have the same slope for candies with chocolate as for candies without. If you want the same slope, that is called the equal slopes model.

```
mod_equal_slopes <- lm(winpercent ~ pricepercent + as.factor(chocolate),
                        data = candy)
summary(mod_equal_slopes)
```

Call:

```
lm(formula = winpercent ~ pricepercent + as.factor(chocolate),
    data = candy)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-25.2821	-7.5946	-0.1665	8.5309	25.1001

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	41.57128	2.40375	17.294	< 2e-16 ***
pricepercent	0.01665	0.05077	0.328	0.744
as.factor(chocolate)1	18.29798	2.90880	6.291	1.46e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.48 on 82 degrees of freedom

Multiple R-squared: 0.4059, Adjusted R-squared: 0.3914

F-statistic: 28.02 on 2 and 82 DF, p-value: 5.337e-10

Different slopes model

If you want to allow the slopes to vary, that is called a different slopes model.

```
mod_diff_slopes <- lm(winpercent ~ pricepercent * as.factor(chocolate),
                        data = candy)
summary(mod_diff_slopes)
```

Call:

```
lm(formula = winpercent ~ pricepercent * as.factor(chocolate),
    data = candy)
```

Residuals:

	Min	1Q	Median	3Q	Max
--	-----	----	--------	----	-----

-26.705 -7.482 -0.658 8.361 24.715

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	44.03751	2.83637	15.526	<2e-16 ***
pricepercent	-0.05525	0.06744	-0.819	0.4150
as.factor(chocolate)1	10.13569	5.85807	1.730	0.0874 .
pricepercent:as.factor(chocolate)1	0.16200	0.10123	1.600	0.1134

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.37 on 81 degrees of freedom

Multiple R-squared: 0.4241, Adjusted R-squared: 0.4028

F-statistic: 19.89 on 3 and 81 DF, p-value: 9.413e-10

Logistic Regression

Consider this model when:

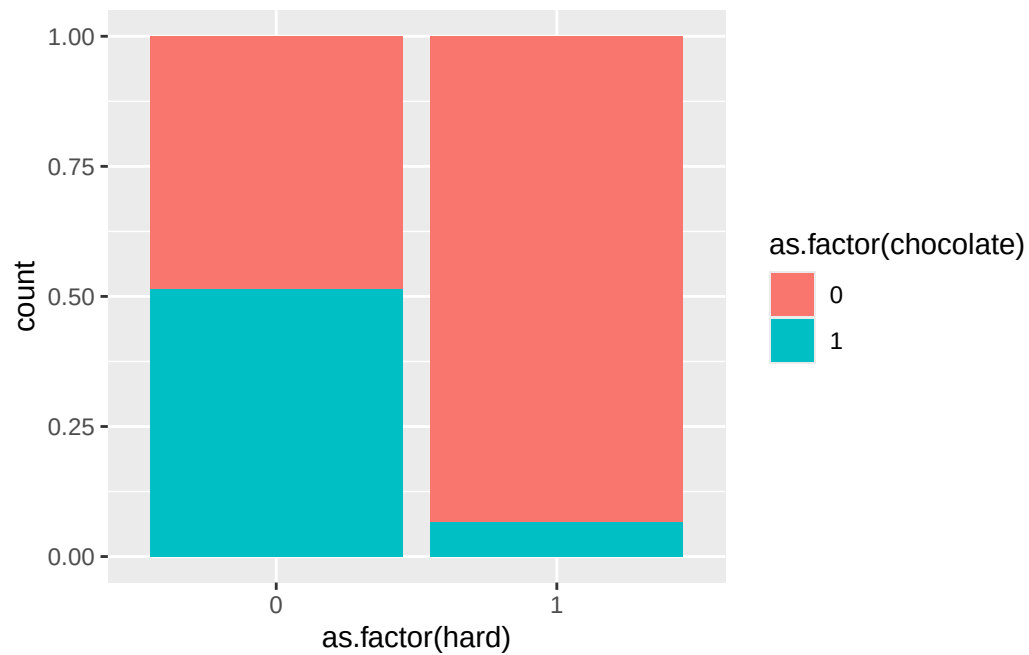
- Response variable (y): categorical (binary)
- Explanatory variable (x): quantitative and/or categorical

Question of interest: Does this candy have chocolate in it??

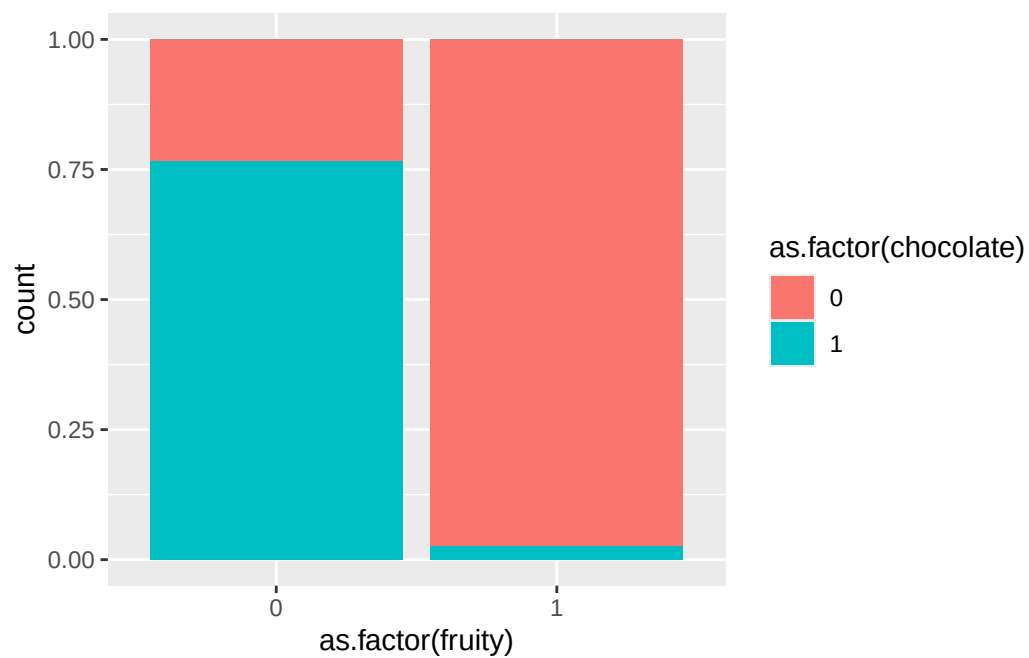
Follow-up question: Which variables in our dataset do you think would be good predictors of whether or not a candy will have chocolate in it?

Visualize the relationships between the variables

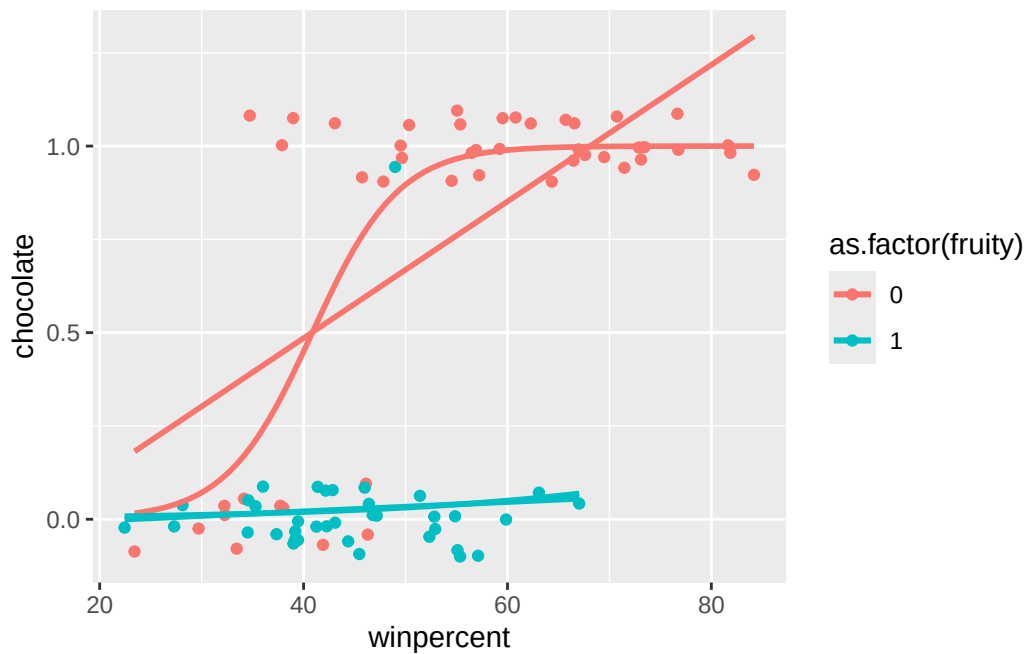
```
# Potential graphs
ggplot(data = candy, mapping = aes(x = as.factor(hard),
                                   fill = as.factor(chocolate))) +
  geom_bar(position = "fill")
```



```
ggplot(data = candy, mapping = aes(x = as.factor(fruity),  
                                   fill = as.factor(chocolate))) +  
  geom_bar(position = "fill")
```




```
ggplot(data = candy, mapping = aes(x = winpercent,
                                   y = chocolate,
                                   color = as.factor(fruity))) +
  geom_jitter(width = 0, height = 0.1) +
  geom_smooth(method = "lm", se = FALSE) +
  geom_smooth(method = "glm", method.args = list(family = "binomial"),
              se = FALSE)
```



Build a logistic regression model

```
# Linear model
mod_lin <- lm(chocolate ~ as.factor(fruity) + winpercent, data = candy)
summary(mod_lin)
```

Call:

```
lm(formula = chocolate ~ as.factor(fruity) + winpercent, data = candy)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.63922	-0.14170	0.01374	0.17507	0.90544

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.010511	0.129843	-0.081	0.936
as.factor(fruity)1	-0.582356	0.065592	-8.879	1.26e-13 ***
winpercent	0.014034	0.002229	6.295	1.44e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.278 on 82 degrees of freedom

Multiple R-squared: 0.6967, Adjusted R-squared: 0.6893

F-statistic: 94.18 on 2 and 82 DF, p-value: < 2.2e-16

```
# Logistic model
mod_log <- glm(chocolate ~ as.factor(fruity) + winpercent, data = candy,
              family = "binomial")
summary(mod_log)
```

Call:

```
glm(formula = chocolate ~ as.factor(fruity) + winpercent, family = "binomial",
    data = candy)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-7.62478	2.40682	-3.168	0.001535 **
as.factor(fruity)1	-5.79549	1.48680	-3.898	9.7e-05 ***
winpercent	0.18799	0.05552	3.386	0.000709 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 116.407 on 84 degrees of freedom

Residual deviance: 32.104 on 82 degrees of freedom

AIC: 38.104

Number of Fisher Scoring iterations: 7

Prediction

You can use the `predict()` function in the same way as we did for linear regression.

```
new_cases <- data.frame(fruity = 0, winpercent = 50)
predict(mod_log, newdata = new_cases, type = "response")
```

```
1
0.8550401
```

Classification Trees

Consider this model when:

- Response variable (y): categorical
- Explanatory variable (x): quantitative and/or categorical

This is a more flexible model fit than logistic regression.

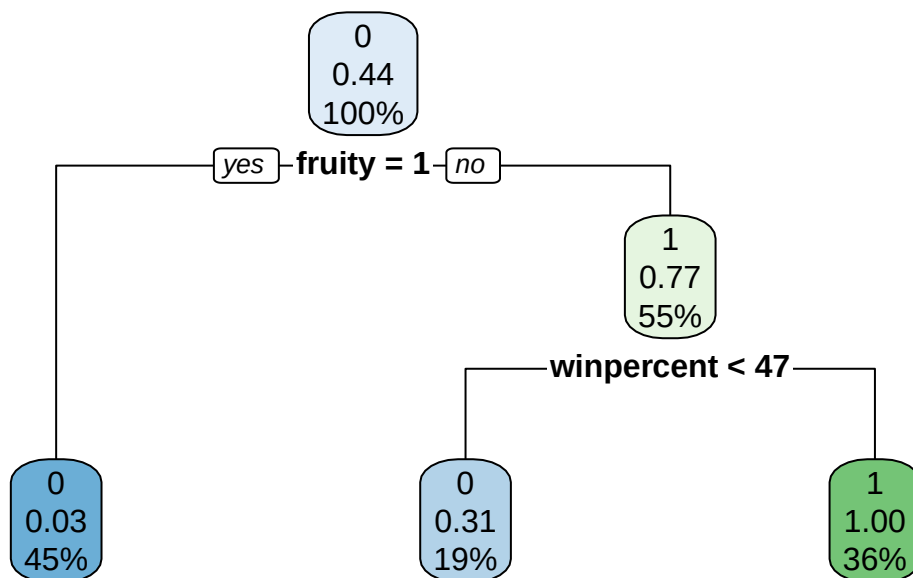
```
# Fit the model
tree <- rpart(chocolate ~ . - competitorname, data = candy, method = "class")
tree
```

```
n= 85
```

```
node), split, n, loss, yval, (yprob)
      * denotes terminal node
```

```
1) root 85 37 0 (0.56470588 0.43529412)
  2) fruity>=0.5 38 1 0 (0.97368421 0.02631579) *
  3) fruity< 0.5 47 11 1 (0.23404255 0.76595745)
    6) winpercent< 47.06318 16 5 0 (0.68750000 0.31250000) *
    7) winpercent>=47.06318 31 0 1 (0.00000000 1.00000000) *
```

```
rpart.plot(tree)
```



Which model is better: the logistic regression model or the classification tree?

Three metrics:

- Accuracy: Proportion of the guesses were correct
- Sensitivity: Among the chocolate candies, proportion of times that the model identified the candy has chocolate
- Specificity: Among the non-chocolate candies, proportion of times that the model identified the candy doesn't have chocolate

Let's draw a confusion matrix on the board.

```
# How do we access the predictions?
mod_log$fitted.values
```

	1	2	3	4	5	6
0.9930717593	0.9938421950	0.1736410990	0.7397432702	0.0271030505	0.8629517717	
7	8	9	10	11	12	
0.9558307979	0.0383285038	0.3824594355	0.0009757615	0.4260742583	0.0012931995	
13	14	15	16	17	18	
0.0001492352	0.0041786848	0.0024674304	0.0048688710	0.0023433814	0.0097036752	
19	20	21	22	23	24	
0.0640219308	0.2308960271	0.0228591193	0.0041063273	0.9418718857	0.9834396396	
25	26	27	28	29	30	
0.9523403317	0.9709987112	0.0002937128	0.9581869379	0.9988951129	0.0035421407	
31	32	33	34	35	36	

```

0.0023239271 0.0300759597 0.9970112146 0.9925387186 0.0090548161 0.9385829224
      37      38      39      40      41      42
0.9978001523 0.9782278139 0.9887154063 0.7968476718 0.9324921076 0.0467840884
      43      44      45      46      47      48
0.9965734663 0.9923926640 0.0001009495 0.0024610736 0.7462081297 0.9956682326
      49      50      51      52      53      54
0.3697286677 0.0034607785 0.0016602131 0.9995759405 0.9997254840 0.9979343096
      55      56      57      58      59      60
0.9977110625 0.0011282190 0.9912437059 0.1149850852 0.0046552731 0.2502249957
      61      62      63      64      65      66
0.1735126952 0.0447294856 0.3769790426 0.0083795584 0.9988752643 0.9725108551
      67      68      69      70      71      72
0.1028017327 0.0296109068 0.3062080104 0.0009870617 0.2076949961 0.1728309046
      73      74      75      76      77      78
0.0002515951 0.0428208726 0.0145996040 0.6157899065 0.7257656829 0.8467783120
      79      80      81      82      83      84
0.0104339048 0.9995577649 0.0075915868 0.0022683073 0.0061928604 0.5628620667
      85
0.8435957161

```

```
predict(tree, type = "class")
```

```

 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
1  1  0  0  0  1  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  1  1  1
27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52
0  1  1  0  0  0  1  1  0  1  1  1  1  1  1  0  1  1  0  0  0  1  0  0  0  1
53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78
1  1  1  0  1  0  0  0  0  0  0  0  0  1  1  0  0  0  0  0  0  0  0  0  0  1
79 80 81 82 83 84 85
0  1  0  0  0  0  1
Levels: 0 1

```

```

# Add predictions
candy <- candy %>%
  mutate(log_model_guess = if_else(mod_log$fitted.values >= 0.5, 1, 0),
         tree_model_guess = predict(tree, type = "class"))

# Confusion matrix
count(candy, chocolate, log_model_guess)

```

```
# A tibble: 4 x 3
```

	chocolate	log_model_guess	n
	<dbl>	<dbl>	<int>
1	0	0	45
2	0	1	3
3	1	0	4
4	1	1	33

```
count(candy, chocolate, tree_model_guess)
```

```
# A tibble: 3 x 3
  chocolate tree_model_guess    n
    <dbl> <fct>          <int>
1       0 0             48
2       1 0             6
3       1 1            31
```

Resources related to modeling

We have just scratched the surface of building and interpreting models in R. Here are some additional resources on modeling:

- Chapters 5, 6, and 10 of [ModernDive](#)
- For a more machine learning approach, check out [Tidy Modeling with R](#) or [Introduction to Statistical Learning in R](#)
- For more advanced modeling, check out [Beyond Multiple Linear Regression](#)
- For bayesian model, check out [Bayes Rules!](#)

Homework

- Download materials from Posit Cloud.
- Go back over all the materials we have covered and try to apply to your own work.
- Bring ideas to the weekly **Writing/Coding Sessions**: 2:00 - 4:00pm on Mondays in June & July, Taylor Hall 203.